Robotics and Intelligent Systems

S5 PCADT502 – 2024 Scheme

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Module 1 Notes

Exhaustive Study Material

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1 Introduction to Robotics

Robotics is a multidisciplinary field of engineering and science that involves the design, construction, operation, and use of robots, as well as computer systems for their control, sensory feedback, and information processing. The primary objective of robotics is to create intelligent machines that can assist humans in a variety of ways. According to the Robot Institute of America (RIA), a robot is defined as a "reprogrammable, multifunctional manipulator designed to move material, parts, tools, or specialized devices through various programmed motions for the performance of a variety of tasks."

1.1 Historical Evolution of Robotics

The term 'robot' originates from the Czech word 'robota', meaning forced labor, introduced by Karel Čapek in his 1920 play *R.U.R. (Rossum's Universal Robots)*. However, the true industrial robotics revolution began in 1954 when George Devol patented the first programmable, digital robot. In 1961, Unimation (founded by Devol and Joseph Engelberger, universally recognized as the 'Father of Robotics') deployed the Unimate robot at a General Motors plant. Its task was to extract hot, heavy metal parts from a die-casting machine—a quintessential 'dangerous and dirty' job. This marked the transition of robots from science fiction into practical, industrial heavy-duty tools. Over the decades, robotics evolved through purely mechanical automation, to microprocessor-controlled precision machines (like the Stanford Arm and the PUMA arm), and finally to today's AI-driven, sensor-rich collaborative autonomous systems.

1.2 Laws of Robotics (Expanded Functional View)

While Asimov's laws are theoretical (discussed further in the Ethics section), practical robotic design is governed by functional imperatives:

1. **Reliability and Precision:** The system must perform repetitive tasks with sub-millimeter accuracy over thousands of operational hours without failure.
2. **Environmental Adaptability:** The system must be capable of processing external noise and variations in its workspace without halting operations.
3. **Failsafe Architecture:** In the event of power loss or sensor failure, the robot must default to a state that cannot harm nearby human operators (e.g., brakes automatically engage when power is cut).

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1.3 Degrees of Freedom (DOF)

The concept of Degrees of Freedom (DOF) is fundamental in understanding and designing robotic systems. DOF refers to the number of independent parameters or coordinates that define the configuration or state of a mechanical system completely.

In a purely spatial sense, a free-floating rigid body in three-dimensional space has exactly six degrees of freedom: three translational (movement along the X, Y, and Z axes) and three rotational (pitch, roll, and yaw about these axes). When bodies (links) are connected by joints to form a robotic manipulator, the total degrees of freedom of the system is constrained by the nature of these joints.

Kinematic Definitions

Grübler-Kutzbach Criterion

The mobility or DOF (M) of a kinematic chain can be determined using the Grübler-Kutzbach formula:

$$M = \lambda(N - 1 - J) + \sum_{i=1}^{J} f_i \tag{1}$$

Where:

- M = Mobility or Degrees of Freedom
- λ = Spatial dimensionality factor ($\lambda = 3$ for planar, $\lambda = 6$ for spatial mechanisms)
- N = Total number of links (including the fixed base)
- J = Total number of joints
- f_i = Degrees of freedom of the ith joint

For most industrial manipulators, achieving arbitrary position and orientation of the end-effector in 3D space requires at least 6 DOF. If a robot has more than 6 DOF, it is termed *kinematically redundant*, which is highly beneficial for obstacle avoidance and manipulating objects in constrained environments.

1.4 Robot Types

Robots can be broadly classified based on their application, control systems, and kinematics:

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- **Industrial Robots:** Typically stationary manipulators used in manufacturing for welding, painting, assembly, pick and place, packaging, and inspection.
- **Service Robots:** Designed to assist human beings, typically by performing jobs that are dirty, dull, distant, dangerous, or repetitive, including household chores and medical surgery.
- **Mobile Robots:** Capable of moving around in their environment and are not fixed to one physical location.
- **Cobots (Collaborative Robots):** Designed to share a workspace with humans, featuring advanced sensors to ensure safety during direct physical interaction.

2 Manipulators

An industrial robot manipulator is essentially a mechanical arm constructed of rigid bodies connected by joints. The structure is analogous to the human arm.

2.1 Anatomy of a Robotic Manipulator

The anatomy of a robot refers to its physical construction and the arrangement of its mechanical parts.

Robot Components

Fundamental Components:

- **Base:** The fixed or mobile foundation to which the first link is attached.
- **Links:** The rigid solid mechanical structural members that connect the joints and maintain a fixed relationship between them.
- **Joints:** The movable connections between links that permit relative motion.
- **End Effector:** The device attached to the wrist of the manipulator to perform the specific task (e.g., gripper, welding gun).

2.2 Links and Joints

The manipulator is a sequence of link-joint-link combinations. Each joint provides a specific type of motion and adds to the overall DOF of the system. There are two primary

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types of joints used in almost all industrial robots:

1. **Revolute Joint (R):** Allows rotary (rotational) motion around an axis. It provides 1 DOF. It is analogous to a hinge or a human elbow.

2. **Prismatic Joint (P):** Allows linear (translational) sliding motion along an axis. It provides 1 DOF. It is analogous to a piston or a sliding door.

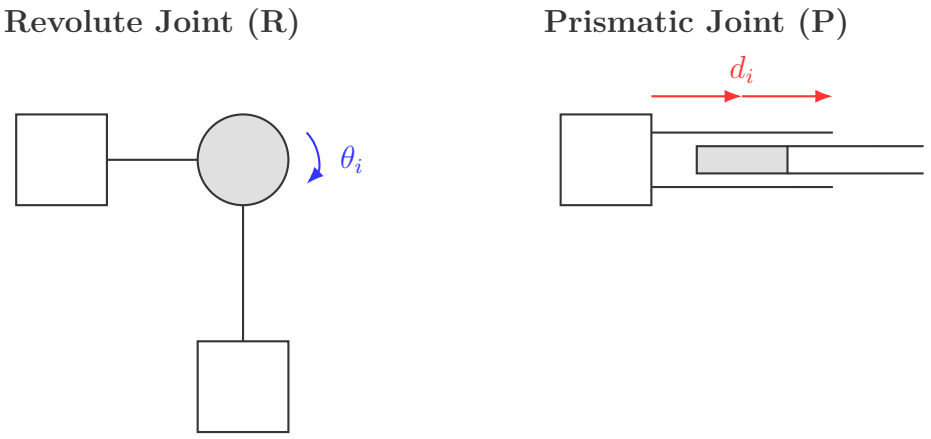


Figure 1: Schematic Representation of Revolute and Prismatic Joints

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2.3 Workspace (Work Envelope)

The workspace (or work volume) of a manipulator is the total physical 3D volume of space that the robot's end-effector can reach. It is determined exclusively by the physical limits of the joints (stroke lengths for prismatic joints, angular limits for revolute joints) and the lengths of the structural links.

- **Reachable Workspace:** The volume of space within which every geometric point can be reached by the end-effector in *at least one* orientation.

- **Dexterous Workspace:** The volume of space within which every geometric point can be reached by the end-effector in *all possible orientations*. Because reaching a point from a highly constrained angle is difficult, the dexterous workspace is mathematically always a subset (a smaller internal volume) of the reachable workspace.

2.4 Denavit-Hartenberg (D-H) Convention

To systematically define the kinematics of a robot arm, engineers globally use the Denavit-Hartenberg (D-H) convention. Instead of relying on complex arbitrary geometry, the D-H convention assigns a local reference coordinate frame to each joint. It describes the

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translation and rotation between adjacent frames ($i-1$ to i) using a transformation matrix dependent on exactly four parameters:

1. **Link Length (a_i):** The distance between the Z_{i-1} and Z_i axes measured along the X_i axis. (Represents the physical length of the link).
2. **Link Twist (α_i):** The angle from the Z_{i-1} axis to the Z_i axis measured about the X_i axis. (Represents the structural twist of the link).
3. **Joint Distance (d_i):** The distance from the X_{i-1} axis to the X_i axis measured along the Z_{i-1} axis. (Variable for prismatic joints).
4. **Joint Angle (θ_i):** The angle from the X_{i-1} axis to the X_i axis measured about the Z_{i-1} axis. (Variable for revolute joints).

D-H Transformation Matrix

The standard transformation matrix T_i^{i-1} mapping frame i to frame $i-1$ is the multiplication of four basic transformations:

$$T_i^{i-1} = Rot(Z,\theta_i) \cdot Trans(Z,d_i) \cdot Trans(X,a_i) \cdot Rot(X,\alpha_i) \quad (2)$$

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2.5 Actuators, Sensors, and Controllers

- **Actuators:** The "muscles" of the robot. They convert stored energy into mechanical work to drive the joints.
  - *Electric Motors:* DC servomotors, AC servomotors, and stepper motors. Highly accurate, clean, and easily controlled.
  - *Hydraulic Actuators:* Use pressurized fluid. Provide very high power-to-weight ratios and large payload capabilities, but are prone to leakage and require bulky infrastructure.
  - *Pneumatic Actuators:* Use compressed air. Fast, inexpensive, and clean, but highly non-linear and difficult to control precisely due to air compressibility.
- **Sensors:** The "senses" of the robot.
  - *Proprioceptive sensors:* Measure internal states like joint position (encoders, potentiometers), velocity (tachometers), and torque.

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  - *Exteroceptive sensors:* Measure the external environment (vision systems, laser range finders, tactile sensors, ultrasonic sensors).

- **Controllers:** The "brain". It is a microprocessor-based system that processes sensory inputs, computes the necessary kinematics/dynamics algorithms, and sends control signals to the actuators to achieve the desired motion.

2.5.1 In-Depth Look at Actuators

- **DC Servomotors:** The gold standard for industrial robotics. They utilize an internal closed-loop feedback mechanism (usually an optical encoder) to correct positional errors continuously. The output torque τ is directly proportional to the armature current I_a, governed by the equation $\tau = K_t I_a$, where K_t is the motor torque constant.

- **Stepper Motors:** Used in low-cost or open-loop systems. They move in discrete angular steps (e.g., 1.8° per step). While they do not require complex feedback loops, they are prone to "skipping" steps under high load, causing catastrophic position loss.

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2.5.2 In-Depth Look at Sensors

- **Absolute vs. Incremental Encoders:** Incremental encoders generate pulses as the motor turns, requiring the robot to perform a "homing" sequence upon startup to find its zero position. Absolute encoders read a unique digital code from an optical disk for every position, allowing the robot to know its exact joint angle instantly upon boot-up without homing.

- **Tactile Sensors:** Advanced end-effectors use piezoelectric or piezoresistive arrays to detect slip, texture, and grip force. This mimics human skin and prevents the robot from crushing delicate items like eggs or fruit.

- **Machine Vision:** Using high-speed cameras coupled with Convolutional Neural Networks (CNNs), modern robots can identify randomly oriented parts on a moving conveyor belt, calculate their center of mass, and adjust the end-effector trajectory dynamically.

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3 Robot Configurations

The combination and arrangement of prismatic (P) and revolute (R) joints determine the kinematic configuration of a manipulator arm. The primary configurations dictate the shape of the robot's workspace—the spatial volume that the robot's end-effector can reach.

3.1 Cartesian Configuration (PPP)

A Cartesian robot consists of three prismatic joints whose axes are coincident with a Cartesian coordinate system (X, Y, and Z).

- **Kinematics:** Highly straightforward decoupled kinematics. Position (x, y, z) corresponds directly to joint displacements (d_1, d_2, d_3).
- **Workspace:** A rectangular box or cube.
- **Advantages:** High accuracy, rigidity, high payload capacity throughout the workspace.
- **Applications:** 3D printing, CNC machining, automated storage and retrieval systems, heavy pick-and-place.

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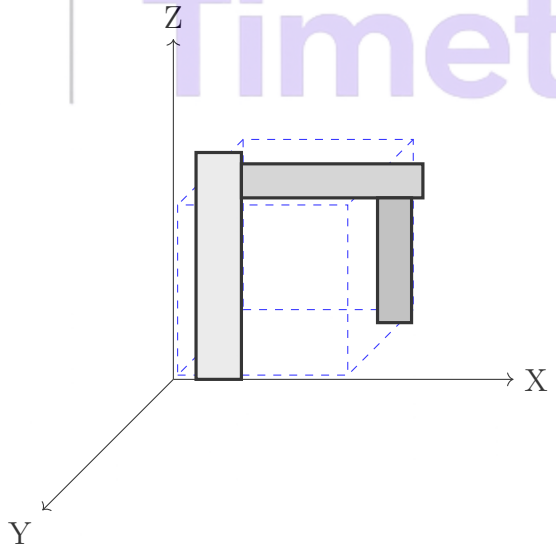


Figure 2: Cartesian (PPP) Configuration and Rectangular Workspace

3.2 Cylindrical Configuration (RPP)

Consists of one revolute joint at the base and two prismatic joints.

- **Kinematics:** Described by cylindrical coordinates (θ, r, z).

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- **Workspace:** A hollow cylinder. The robot cannot reach its own central axis due to the mechanical structure.
- **Applications:** Assembly operations, machine tending, handling die-cast machines.

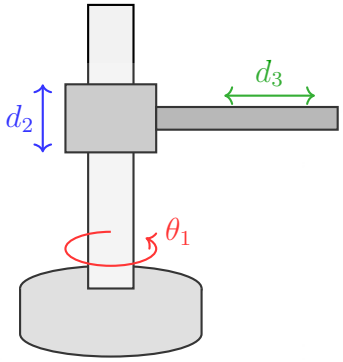


Figure 3: Cylindrical (RPP) Configuration Kinematic Diagram

3.3 Spherical Configuration (RRP)

Also known as polar configuration, it consists of two revolute joints and one prismatic joint.

- **Kinematics:** Based on spherical coordinates (θ, ϕ, r).
- **Workspace:** A spherical envelope (specifically, a segment of a sphere).
- **Applications:** Historically significant (the first industrial robot, Unimate, was RRP). Used in spot welding and fettling machines.

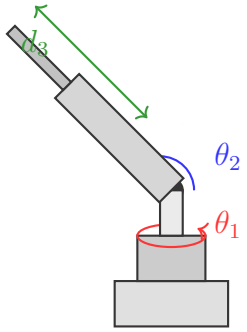


Figure 4: Spherical (RRP) Configuration Kinematic Diagram

3.4 Articulated Configuration (RRR)

Also known as jointed-arm configuration, consisting of three revolute joints (waist, shoulder, and elbow), closely mimicking the human arm.

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- **Kinematics:** Complex, highly coupled nonlinear kinematics. Requires intense computational power for inverse kinematics.
- **Workspace:** A complex, roughly spherical shape. Offers the highest reachability and dexterity among all configurations.
- **Advantages:** Maximum flexibility, can reach over and under obstacles.
- **Applications:** Arc welding, painting, advanced assembly, automotive manufacturing.

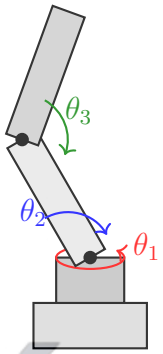


Figure 5: Articulated (RRR) Configuration Kinematic Diagram

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4 Mobile Robots

Unlike fixed industrial manipulators, mobile robots have the capability to move through their environment. They are classified based on their locomotion mechanisms.

4.1 Wheeled Mobile Robots

Wheeled robots are the most common type of mobile robots due to their high efficiency on smooth, structured surfaces. They suffer on extremely rugged terrain.

- **Differential Drive:** Features two independently driven wheels on a common axis. The robot changes direction by varying the relative rate of rotation of its wheels.

Kinematic Definitions

Differential Kinematics:

$$v = \frac{r(\omega_R + \omega_L)}{2} \tag{3}$$

$$\omega = \frac{r(\omega_R - \omega_L)}{L} \tag{4}$$

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Where v is linear velocity, ω is angular velocity, r is wheel radius, ω_R, ω_L are wheel angular speeds, and L is the track width.

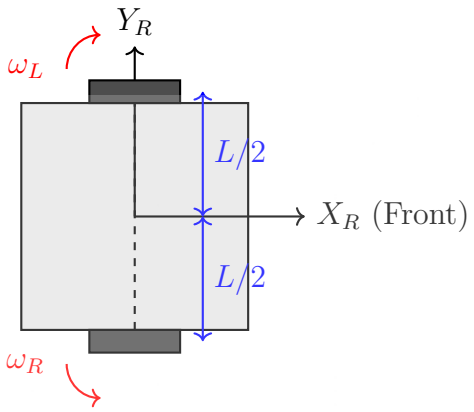


Figure 6: Differential Drive Mobile Robot Layout

- **Tricycle Drive:** One steered and driven wheel, with two passive wheels.
- **Ackermann Steering:** Standard car steering mechanism where the inner wheel turns at a sharper angle than the outer wheel to prevent slipping during a turn.
- **Omnidirectional (Mecanum/Holonomic):** Uses special wheels with rollers set at angles, allowing the robot to move in any direction (X, Y, and rotation) instantaneously without turning.

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4.1.1 Odometry and Dead Reckoning in Wheeled Robots

Odometry is the fundamental methodology used by wheeled mobile robots to estimate their current position relative to a starting location. For a differential drive robot, wheel encoders measure the exact fractional rotation of the left and right wheels. By integrating these discrete velocity measurements over tiny time intervals (Δt), the robot's state vector (Pose) $[x, y, \theta]^T$ is updated continuously.

Odometry State Update Equations

Assuming the robot travels at linear velocity $V(t)$ and angular velocity $\omega(t)$:

$$x(t) = x_0 + \int_0^t V(t)\cos(\theta(t))dt \tag{5}$$

$$y(t) = y_0 + \int_0^t V(t)\sin(\theta(t))dt \tag{6}$$

$$\theta(t) = \theta_0 + \int_0^t \omega(t)dt \tag{7}$$

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While mathematically sound, odometry is highly susceptible to accumulated errors (known as *drift*). If a wheel slips on a smooth floor, the encoder registers movement that did not physically occur. To correct this, robots must fuse odometry data with exteroceptive sensors like LIDAR (Light Detection and Ranging) using complex algorithms like the Kalman Filter or SLAM (Simultaneous Localization and Mapping).

4.1.2 Path Planning Algorithms

For a mobile robot to navigate autonomously, it must compute a collision-free trajectory from point A to point B.

- **Global Path Planning:** Uses a complete map of the environment. Algorithms like **Dijkstra's Algorithm** or **A* (A-Star)** evaluate a grid map to find the mathematically shortest path.
- **Local Path Planning (Obstacle Avoidance):** Uses real-time sensor data to avoid unexpected obstacles. Techniques like the **Artificial Potential Field (APF)** method model the goal as an attractive magnetic force and obstacles as repulsive forces, naturally guiding the robot around humans or dropped boxes.

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4.2 Legged Mobile Robots

Legged robots are inspired by biological systems and provide superior mobility over extremely uneven, rocky, or disconnected terrain where wheels would fail.

- **Bipedal (2 Legs):** Humanoid robots. They are *dynamically stable*, meaning they must actively balance to avoid falling. They utilize the Zero Moment Point (ZMP) concept for stability control.
- **Quadruped (4 Legs):** Dog-like robots (e.g., Boston Dynamics Spot). Capable of dynamic gaits like trotting and galloping.
- **Hexapod (6 Legs):** Insect-like. They are *statically stable* because they can keep three legs on the ground at all times (alternating tripod gait), forming a stable support triangle.

4.3 Aerial, Underwater, and Surface Water Robots

- **Aerial Robots (UAVs):** Unmanned Aerial Vehicles include fixed-wing drones and multi-rotors (quadcopters). Quadcopters use 4 rotors where opposite pairs rotate

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in the same direction, controlling pitch, roll, yaw, and thrust by varying individual motor speeds.

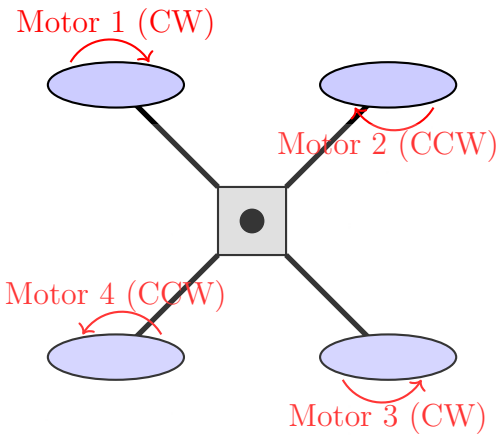


Figure 7: Quadcopter Rotor Configuration showing counter-acting torques. By speeding up Motor 1 and 3 while slowing 2 and 4, the drone Yaws without losing total thrust.

- **Underwater Robots:**
  - *ROV (Remotely Operated Vehicle):* Tethered to a surface vessel, providing power and high-bandwidth communication for deep-sea exploration.
  - *AUV (Autonomous Underwater Vehicle):* Untethered, pre-programmed, carrying their own energy source. Used for mapping ocean floors.
- **Surface Water Robots (USVs):** Unmanned Surface Vehicles operate on the water surface. Used for environmental monitoring, hydrographic surveys, and military surveillance.

5 Dynamic Characteristics

The dynamic characteristics of a robot refer to how the system behaves under the influence of forces, torques, and accelerations over time. This is distinctly different from kinematics, which only studies motion without considering the forces causing it.

5.1 Speed of Motion

Speed of motion defines how fast the robot's end-effector can move from one point to another. It is constrained by the maximum rotational speeds of the actuators, the structural rigidity of the links, and the computational speed of the controller.

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- **Velocity Profiles:** To prevent immense mechanical stress and jerk (rate of change of acceleration), robotic controllers do not demand instantaneous velocity changes. They use trapezoidal or S-curve velocity profiles. In a trapezoidal profile, the robot accelerates linearly to a cruise velocity, maintains it, and then decelerates linearly to a stop.
- **Cycle Time:** In industrial settings, the speed of motion is optimized to minimize cycle time (the time taken to complete one full operational loop, such as picking a part and placing it on a conveyor).

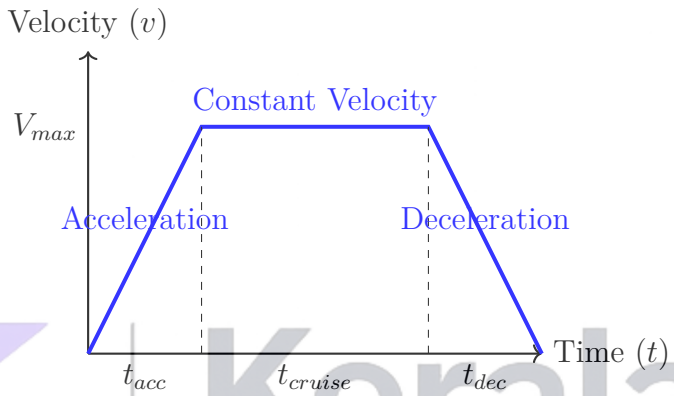


Figure 8: Trapezoidal Velocity Profile for Joint Motion Control

5.2 Advanced Dynamic Modeling and The Jacobian

To accurately control a robot operating at high speeds, the controller must compute the required torques using advanced dynamic models.

- **Lagrange-Euler Formulation:** Based on the kinetic (K) and potential (P) energy of the system. The Lagrangian is defined as $L = K - P$. It provides elegant, closed-form equations but requires $O(n^4)$ mathematical operations, making it highly computationally intensive for high-DOF robots.
- **Newton-Euler Formulation:** Based on Newton's Second Law and Euler's momentum balance equations. It evaluates forces recursively from the base to the end-effector (forward recursion) and torques from the end-effector back to the base (backward recursion). It operates in $O(n)$ time, making it the industry standard for real-time robotic control algorithms.
- **The Jacobian Matrix (J):** A critical mathematical matrix that relates the ve-

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locity of the joints to the linear and angular velocity of the end-effector.

$$V_{end_effector} = J(\theta) \cdot \dot{\theta} \tag{8}$$

When the Jacobian matrix loses rank (its determinant becomes zero), the robot enters a **Kinematic Singularity**. In a singularity, the robot loses one or more degrees of freedom, meaning there are certain directions the end-effector physically cannot move, no matter how fast the joint motors spin.

5.3 PID Control for Speed of Response

The actual speed of response of a robot joint is dictated by its servo drive, which heavily relies on a PID (Proportional-Integral-Derivative) controller.

- **Proportional (K_p):** Applies force proportional to the current positional error. High K_p makes the robot fast but causes it to overshoot the target.
- **Integral (K_i):** Accumulates past errors to eliminate steady-state error (e.g., sagging under heavy load). Too much K_i causes the system to oscillate wildly.
- **Derivative (K_d):** Predicts future error based on the current rate of change. It acts as an electronic damper, smoothing out the motion and preventing the overshoot caused by the Proportional term.

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5.4 Load Carrying Capacity

Load carrying capacity (or payload) is the maximum weight a robot can manipulate continuously at a specified speed without degrading its performance (precision, repeatability, or structural integrity).

- **Rated Payload vs. Maximum Payload:** Rated payload is the mass the robot can handle while operating at maximum speed and maximum reach. The absolute maximum payload might be higher but requires the robot to move slower or operate closer to its base to reduce the lever arm effect.
- **Dynamic Loading:** When a robot accelerates, the payload generates inertial forces ($F = ma$) and moments. The joints must supply enough torque to overcome not only the static weight of the payload due to gravity but also these dynamic inertial forces.

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- **Payload-to-Weight Ratio:** A critical metric in robotics. Traditional industrial robots have low ratios (e.g., a 1000 kg robot lifting 100 kg), meaning they spend most of their energy moving their own mass. Modern lightweight cobots aim for better ratios.

5.5 Speed of Response

Speed of response indicates how quickly the robot's mechanical and control systems can react to a new command or a disturbance.

- **Transient Response:** When a joint is commanded to move to a new angle, it does not do so instantaneously. The response is characterized by:
  - *Rise Time:* The time required to go from 10% to 90% of the final value.
  - *Overshoot:* How far the arm swings past the target before coming back. High overshoot can cause collisions.
  - *Settling Time:* The time it takes for the vibrations to dampen out and the arm to settle within a specific error band around the target.
- **Damping:** The physical and electronic resistance to motion. An underdamped robot will oscillate severely (wobble) when stopping. A critically damped robot will stop perfectly without oscillation in the minimum time. Industrial robots are typically tuned to be slightly overdamped or critically damped to prevent dangerous oscillations of the end-effector.

6 Introduction to End Effectors

The end effector (or End-Of-Arm Tooling, EOAT) is the device at the end of a robotic arm, designed to interact with the environment. It is the "hand" of the robot, translating the positioning capability of the manipulator into productive work.

6.1 Active and Passive Grippers

- **Active Grippers:** Have internal actuators (motors, pneumatic cylinders) to execute the gripping action. They can control the gripping force and adjust to different object sizes.

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- **Passive Grippers:** Do not have internal moving parts for actuation. They rely on the physical motion of the robot arm to engage with the object (e.g., pushing a mechanical latch, or using environmental forces).

6.2 Mechanical Grippers

Mechanical grippers use mechanical fingers to grasp an object.

- **Friction Gripping:** The fingers rely solely on the friction force generated between the finger pads and the object. High gripping force is required, which may crush fragile objects.
- **Encompassing Gripping:** The fingers are shaped to exactly match the contours of the object, physically caging it. It is much more secure and requires less force, but is object-specific.
- **Linkage Mechanisms:** Most mechanical grippers use gear-and-rack, cam, or four-bar linkage mechanisms to convert the linear motion of a pneumatic cylinder into the rotational motion of the fingers.

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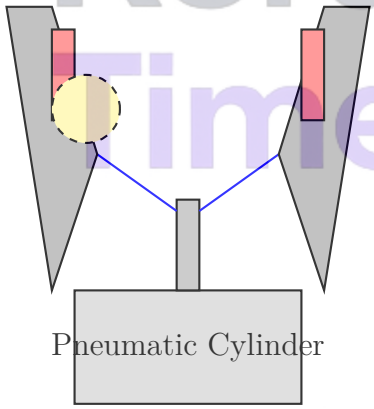


Figure 9: Schematic of a Two-Finger Mechanical Gripper with Linkage

Friction Gripper Force Calculation

For a two-finger friction gripper, the required gripping force F_g to hold a part of mass m against gravity is dependent on the coefficient of friction. To prevent the object from slipping, the total frictional force must be greater than or equal to the gravitational force.

$$F_g = \frac{m \cdot g \cdot F_s}{\mu \cdot n} \tag{9}$$

Where:

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- g = Acceleration due to gravity ($9.81 m/s^2$)
- F_s = Factor of safety (typically 1.5 to 3.0, to account for dynamic acceleration forces)
- μ = Coefficient of static friction between the gripper pads and the object
- n = Number of fingers (usually 2)

6.3 Other Types of Grippers and Special Tools

- **Vacuum Grippers:** Use suction cups to lift objects. A vacuum is generated using an electromechanical pump or a venturi ejector (using compressed air). Excellent for flat, smooth, non-porous surfaces like glass panes, sheet metal, or cardboard boxes. The lifting force is governed by $F = P \times A$, where P is the vacuum pressure and A is the area of the suction cups.

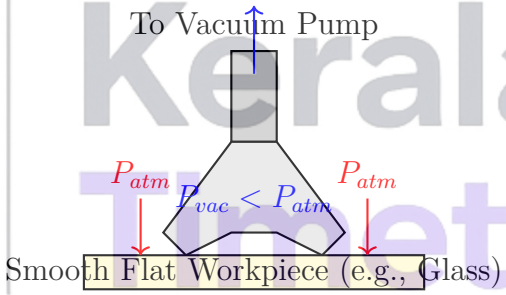


Figure 10: Working Principle of a Vacuum Gripper. The external atmospheric pressure (P_{atm}) physically pushes the object up into the cup because the internal pressure (P_{vac}) has been evacuated.

- **Magnetic Grippers:** Use electromagnets or permanent magnets to pick up ferromagnetic materials. Very fast acting and require access to only one face of the object. Cannot be used if the object must not be magnetized (e.g., CRT screens, certain electronics) or if the material is non-ferrous (aluminum, plastics).
- **Adhesive Grippers:** Use sticky surfaces or advanced biomimetic materials (like synthetic Gecko skin utilizing Van der Waals forces) to adhere to objects. Useful for lightweight fabrics or highly delicate items.
- **Special Tools:** Rather than gripping an object, the robot manipulates a tool. Examples include spot welding guns, arc welding torches, spray painting nozzles, deburring rotary tools, and laser cutting heads.

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7 Ethics in Robotics

As robots become more autonomous and integrated into human society, ethical considerations regarding their design, deployment, and decision-making capabilities are paramount. This field is often referred to as Roboethics.

7.1 Asimov's Three Laws of Robotics

Introduced by science fiction author Isaac Asimov in his 1942 short story "Runaround," these laws serve as the foundational philosophical framework for machine ethics, even though they are fictional and currently impossible to perfectly encode into software.

Core Concept

The Three Laws:

1. **First Law:** A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. **Second Law:** A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.
3. **Third Law:** A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

Zeroth Law (Added later): A robot may not harm humanity, or, by inaction, allow humanity to come to harm.

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While conceptually brilliant, implementing these laws in real-world intelligent systems faces massive hurdles. Defining "harm" mathematically for a machine learning algorithm in a complex, unpredictable physical environment is an unsolved challenge.

7.2 Modern Ethical Considerations

Real-world robotic ethics today focuses on concrete socio-economic and safety issues:

- **Employment and Economics:** The rapid automation of blue-collar and increasingly white-collar jobs poses risks of mass displacement. The ethical question concerns the responsibility of corporations and governments to manage workforce transitions.

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- **Lethal Autonomous Weapons Systems (LAWS):** The development of military drones and robots that can independently identify and engage targets without human intervention raises severe moral and legal questions regarding accountability in warfare.
- **Privacy and Surveillance:** Mobile and service robots (like cleaning robots or domestic assistants) are equipped with high-resolution cameras, microphones, and continuous internet connections, creating massive vulnerabilities regarding domestic privacy.
- **Accountability and Liability:** If an autonomous vehicle or a medical surgical robot makes a fatal error, the ethical and legal frameworks struggle to assign blame. Is it the fault of the hardware manufacturer, the software programmer, the user, or the AI itself?

7.3 The Trolley Problem in Autonomous Systems

A classic ethical dilemma increasingly relevant in robotics is the "Trolley Problem." When an autonomous mobile robot (like a self-driving car or an automated factory forklift) faces an unavoidable crash due to sudden sensor failure or unpredictable human action, how should its algorithms prioritize life? Should it swerve to avoid a crowd of pedestrians, even if it means hitting a solid barrier and fatally injuring its own passenger?

Unlike human drivers who react on split-second instinct, an AI makes calculated, deterministic decisions. Therefore, encoding these moral weightings into a decision matrix implies a pre-meditated calculation of the value of human life. This remains highly controversial and varies drastically across different ethical frameworks (utilitarianism vs. deontological ethics) and cultural contexts.

7.4 Safety Standards (ISO 10218 and ISO/TS 15066)

To mitigate ethical and physical risks in industrial environments, international standards dictate robot deployments.

- **ISO 10218:** Dictates safety requirements for traditional industrial robots, heavily emphasizing heavy metal guarding, light curtains, emergency stops, and completely restricting human access to the active workspace.

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- **ISO/TS 15066:** Specifically addresses Collaborative Robots (Cobots). Because cobots share workspaces with humans without cages, this standard outlines strict biomechanical limits. It defines exactly how much force, pressure, or kinetic energy a robot is legally allowed to exert on different parts of the human body (e.g., hands vs. face) during an accidental, yet inevitable, collision.

7.5 Applications of Robots

Robotics has permeated almost every industrial sector:

- **Manufacturing:** Automotive assembly lines, electronics manufacturing, packaging, and logistics.
- **Healthcare:** Minimally invasive surgical robots (e.g., Da Vinci system), exoskeleton rehabilitation robots, and hospital delivery bots.
- **Space Exploration:** Mars rovers (Curiosity, Perseverance), robotic arms on the ISS (Canadarm).
- **Agriculture:** Automated harvesting, precision spraying drones, and robotic weeders.
- **Disaster Response:** Search and rescue robots capable of entering collapsed buildings or highly radioactive zones where human entry is fatal.

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